

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

9744/02

Paper 2 Structured Questions

September 2017

Additional Materials: Answer Paper

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions in the spaces provided on the Question Paper

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	14
2	7
3	9
4	11
5	9
6	9
7	12
8	9
9	10
10	10
Total	100

This document consists of **5 printed pages and 1 blank page.**

[Turn over

Answer **all** the questions in this section.

1 Table 1.1 shows some features of four biological molecules that are all polymers.

(a) Complete Table 1.1 by using a tick (✓) to indicate the features that apply to each polymer.

Table 1.1

feature	amylopectin	cellulose	RNA	polypeptide
synthesised from amino acid monomers				
contains glycosidic bonds				
polymer is branched				
contains nitrogen				
can be found in both animal and plant cells				

[4]

feature	amylopectin	cellulose	RNA	polypeptide
synthesised from amino acid monomers				✓
contains glycosidic bonds	✓	✓		
polymer is branched	✓			
contains nitrogen			✓	✓
can be found in both animal and plant cells	;	;	✓ ;	✓ ;

(b) Fig. 1.1 is a simple diagram of a phospholipid molecule.

Explain how the structure of a phospholipid molecule makes it suitable for its function in cell membranes. You may label and annotate Fig. 1.1 as part of your answer.



Fig. 1.1

Hydrophilic / polar, phosphate, head / group faces aqueous medium and hydrophobic / non polar, hydrocarbon / fatty acid, tails / chains faces each other / interior of the cell membrane; **R** if labelled correctly but incorrectly described in the text

Forming the phospholipid bilayer resulting in partially permeability / ability to act as a barrier to, hydrophilic substances / water soluble substances / polar substances / ions / AW ;

Presence of unsaturated hydrocarbon tails results in kinks, preventing the close packing of phospholipids, thus regulating fluidity of membrane;

[3]

(c) State two components of a cell surface membrane other than phospholipid molecules and describe their function.

*max two components, one mark each
one mark for function to match the stated component*

Glycolipid / glycoprotein;
Receptors for cell signalling / cell-cell recognition / cell-cell adhesion;

Cholesterol;
Regulate membrane fluidity;

Protein;
Receptor for cell signalling / enzyme / channel protein / carrier protein for facilitated diffusion / active transport

[4]

The fluid mosaic model of membrane structure was first proposed in 1972 by Singer and Nicolson. The model describes in detail how the components of a membrane are organised.

- (d)** Suggest why 'fluid mosaic' is an appropriate term to use to describe membrane structure.

Fluid refers to phospholipids and proteins being able to move freely within the phospholipid bilayer;

Phospholipids are also being able to move laterally within a layer;

Mosaic – protein molecules are embedded and scattered among the phospholipids;

[3]

[Total: 14]

- 2 Erythropoietin, also known as EPO, is a large glycoprotein synthesised by specialised cells in the kidney. These cells are very sensitive to changes in oxygen concentration in the blood passing through the kidney and respond to a low oxygen concentration by increasing the synthesis of EPO.

EPO acts at the surface of particular target cells, such as cells in the bone marrow. These bone marrow cells are stimulated to produce red blood cells.

- (a) A low oxygen concentration leads to an increase in the quantity of mRNA in the specialised cells in the kidney by activating hypoxia-inducing factors (HIFs) which regulates gene expression at the transcriptional level.

Explain how the activation of HIFs results in an increase in EPO mRNA in the cells.

HIFs are activators which bind to enhancer;
Cause DNA to loop through protein-protein interaction and stabilizes the transcription initiation complex;
Enhance binding of RNA polymerase to promoter to increase rate of transcription;

[3]

- (b) All cells of the body are exposed to circulating blood plasma containing EPO, but only particular target cells respond.

- (i) Suggest why EPO acts on target cells and not other cells.

Target cells have cell surface receptors specific for EPO;
Binding sites of receptors are complementary in shape to EPO;

[2]

- (ii) EPO cannot pass through the cell surface membrane to enter the bone marrow cells. Suggest **one** reason why this is so.

Large protein, therefore cannot pass through the gaps between phospholipids;
Hydrophilic molecule, therefore is repelled / cannot pass through the hydrophobic core of the phospholipid bilayer;
No specific membrane transport protein to transport EPO;

[1]

- (c) Red blood cells originate from undifferentiated cells in the bone marrow that are capable of continuous mitotic cell division.

State the name of this type of undifferentiated cell.

stem cell ; A haematopoietic stem cell
treat as neutral adult / non-embryonic / multipotent / stromal

[1]

[Total: 7]

- 3 (a) Fig. 3.1 is a diagram of an ATP molecule.

Label Fig. 3.1 to show the components of an ATP molecule.

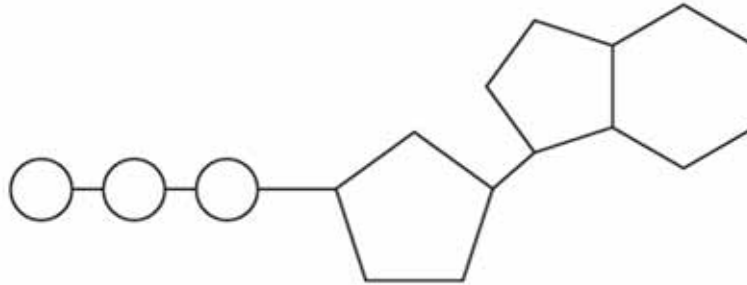


Fig. 3.1

phosphate ;
ribose / pentose sugar ; R deoxyribose
adenine;

labels pointing to correct components

[3]

- (b) ATP is used during translation in amino acid activation, when an amino acid becomes attached to its specific tRNA molecule having a particular anticodon. The reaction requires an enzyme called aminoacyl tRNA synthetase.

- (i) Explain why a particular amino acid needs to be linked to a specific tRNA molecule.

Anticodon on tRNA binds to codon on mRNA via complementary base pairing;
In order to bring the correct amino acid to the ribosome for translation / give correct sequence of amino acids;
tRNAs bring amino acids, adjacent to each other for peptide bond formation;

[2]

- (ii) Explain how the structure of an enzyme such as aminoacyl tRNA synthetase would be altered if the pH of the cytoplasm became too acidic.

Ionic bonds and hydrogen bonds between R groups are disrupted;
Distorts specific 3-dimensional conformation of protein and hence active site is distorted, resulting in denaturation;

[2]

- (iii) Aminoacyl tRNA synthetase uses the induced fit mechanism.

Explain the induced fit mechanism.

Active site is not perfectly complementary / not precise fit to substrate (amino acid and tRNA);

Upon binding / formation of bonds with substrate, enzyme changes conformation and active site becomes perfectly complementary / precise fit to substrate;

[2]

[Total: 9]

- 4 *Morbillivirus* causes measles. The structure of *Morbillivirus* is shown in **Fig. 4.1**.

Haemagglutinin (H) and fusion protein (F) are glycoproteins embedded in the viral envelope.

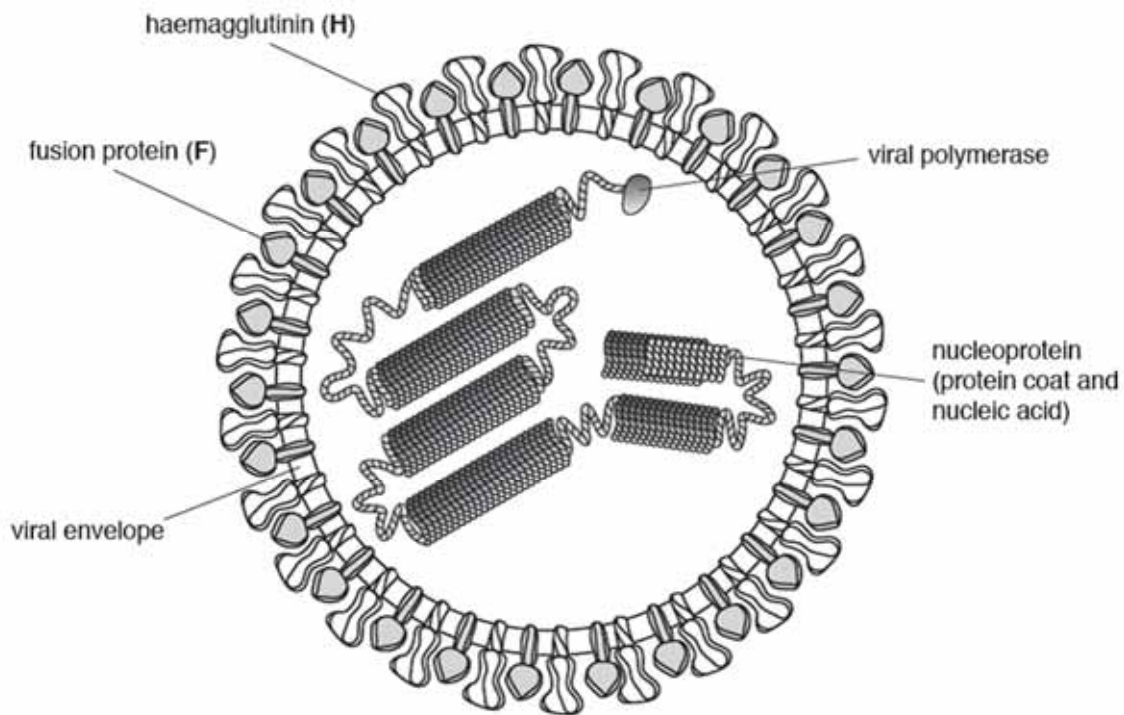


Fig. 4.1

Morbillivirus only infects cells that have a membrane glycoprotein known as signalling lymphocyte activation molecule (SLAM).

When *Morbillivirus* infects a cell, **H** acts before **F**. After the virus binds to the host cell, only the nucleoprotein with the viral polymerase enters the host cell and the virus is replicated.

Morbillivirus replicates its genetic material in the same manner as influenza virus.

New viral particles leave the host cell by budding from the cell surface membrane of the cell. This forms the main part of their envelope.

(a) List two ways in which the structure of *Morbillivirus*:

(i) is similar to HIV

Both have viral envelope (studded with glycoprotein);

Both contains RNA as nucleic acid;

Both contain fusion protein on the viral envelope;

[2]

(ii) differs from the HIV

1 copy of genome vs 2 copies of genome;

Negative sense RNA genome vs Positive sense RNA genome

viral polymerase vs protease + reverse transcriptase + integrase

haemagglutinin vs gp120 embedded in viral envelope;

conical capsid / enclose entire genome vs nucleoprotein

[2]

(b) With reference to **Fig. 4.1** and the information provided,

(i) suggest how *Morbillivirus* infects a cell with SLAM glycoproteins so that only nucleoprotein and viral polymerase enter.

haemagglutinin / H / (viral) glycoprotein, binds to / fits into / complementary to, SLAM / receptor ;

viral envelope fuses with cell surface membrane;

[2]

(ii) suggest the role of viral polymerase in *Morbillivirus*.

replication of RNA / to make copies of genes / AW ;

convert negative sense RNA to positive sense RNA / production of mRNA ;

detail ; e.g. to make viral proteins ;

[3]

(c) Measles has only one serotype. Within this serotype, there are 24 genotypes recognised to date.

Describe how these 24 genotypes could have arisen.

by antigenic drift / mutation

RNA genome mutation due to lack of proof reading by viral polymerase

(Change in haemagglutinin molecule greater infectivity)

[2]

[Total: 11]

- 5 An *Allium cepa* root was cut into ten transverse sections at different distances from the tip. The sections were stained and viewed under the microscope. The number of cells in mitosis were counted in each section and the results were used to determine the mitotic index.

This is calculated as follows:

$$\text{mitotic index} = \frac{\text{number of cells in mitosis}}{\text{total number of cells}}$$

Fig. 5.1 shows the mitotic index for the ten sections.

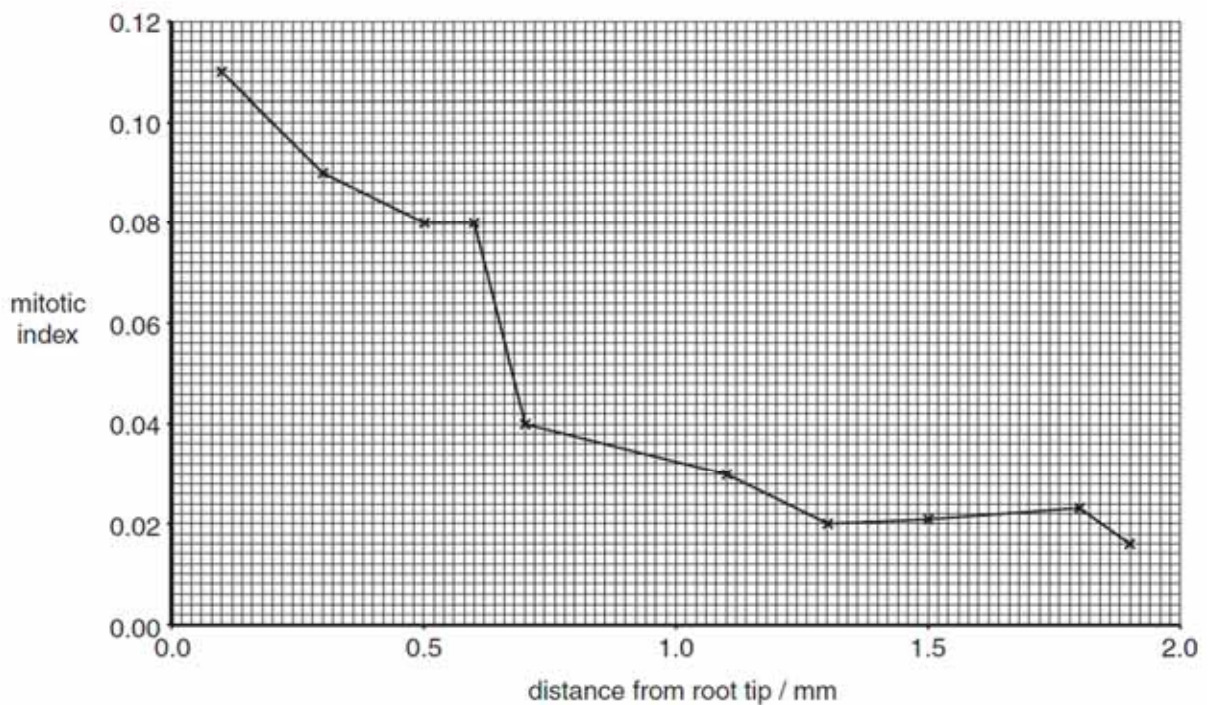


Fig. 5.1

- (a) Using the information in **Fig. 5.1**, describe how the mitotic index changes along the length of the root.

MI decrease from 0.11 to 0.016, as distance from tip increases/from 0.1 to 1.9mm;

Other description with ref to quoted data;

[2]

- (b) Explain how the events in the mitotic cell cycle ensure that all the cells in the root are genetically identical.

semi-conservative DNA replication during, interphase/S phase;

During metaphase, centromeres of chromosomes align at the metaphase plate;

During anaphase, genetically identical sister chromatids are separated/move to opposite poles/go into

separate cells";

After cytokinesis, new cells have same number, and kind of chromosomes/AW e.g. same, genes/DNA/chromosomes as parents;

[3]

- (c) 2 genes, **A/a** and **B/b** are linked on the same pair of homologous chromosomes in *Allium cepa*, as shown in Fig. 5.2.

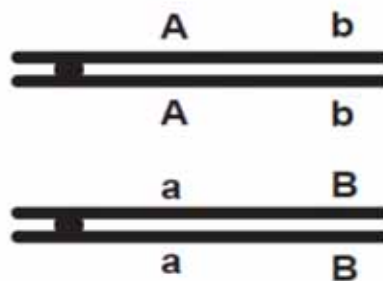
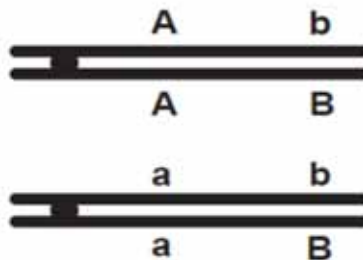


Fig. 5.2

With reference to Fig. 5.2,

- (i) draw a diagram to show the effect of crossing-over between the homologous chromosomes.



Cross over in between two gene loci;

Outcome;

[2]

- (ii) state the effect of linkage and crossing-over on the proportions of gametes with different genotypes that are produced.

large number of, parental types / Ab and aB and small number of, recombinant types / AB and ab;

more recombinants further loci are apart / ora;

[2]

[Total: 9]

- 6 *CFTR*, the cystic fibrosis (CF) gene, encodes for the CFTR protein that plays an essential role in anion regulation and tissue homeostasis of various epithelia. In the gastrointestinal (GI) tract CFTR promotes chloride and bicarbonate secretion, playing an essential role in ion and acid-base homeostasis.

Gene mutations responsible for cystic fibrosis can be classified into 6 different classes.

- **Class I** mutations can result from nonsense and frame-shift mutations, as well as mRNA splicing defects.
- **Class III and IV** mutations are typified by aberrant channel function, rather than reduced quantities of CFTR.

- (a) Explain how **Class I** mutations result in a reduction in the quantity of functional CFTR protein.

Nonsense mutation results in a truncated protein being formed, which will not be able to fold into a functional protein ;

Frame-shift mutation results in a extensive change in amino acid sequence, resulting in a non-functional protein being formed ;

mRNA splicing defects will not result in a mature mRNA formed/does not have the correct continuous coding sequence, hence resulting in a non-functional protein being formed ;

Only one normal allele coding for functional protein instead of two;

[max 2 marks]

[2]

- (b) Explain how **Class III and IV** mutations result in aberrant channel function.

Substitution of a single nucleotide in the CFTR gene, results in change in amino acid in the polypeptide chain;

Different interactions between different R groups, result in change in specific 3-dimensional conformation of protein, hence loss of function;

[2]

CFTR, has been identified as a candidate driver gene for colorectal cancer (CRC) in mice and humans. Further, recent epidemiological and clinical studies indicate that CF patients are at high risk for developing tumours in the colon. Investigations suggest that *CFTR* is a **tumour suppressor gene** in the intestinal tract as *CFTR* mutant mice developed significantly more tumours in the colon and the entire small intestine.

- (c) Explain why a mutation in the tumour suppressor gene, such as the *CFTR* gene, may lead to significantly more tumours in mice.

ref to normal function of TSG;

LOF mutation;

In both alleles of the gene such that no functional protein is produced;

resulted in loss of control of cell cycle/ uncontrolled proliferation of cells leading to tumour formation;

[3]

- (d) Suggest how tumours can eventually become malignant, leading to cancer.

ref to angiogenesis/ blood vessels supplying nutrients and oxygen to tumour;

ref to tissue invasion / metastasis/ loss of density dependent inhibition anchorage thus cells break away to establish themselves in other organs/ tissues;

[2]

[Total: 9]

- 7 (a) Sometimes a gene has more than two alleles, termed *multiple alleles*.

The ABO blood group system in humans is controlled by a gene with three alleles, I^A , I^B and I^O . Alleles I^A and I^B are codominant and I^O is recessive to both.

The blood group **AB** is the result of codominance.

Explain what is meant by *codominance* in this context.

I^A allele codes for A antigen and I^B allele codes for B antigen;

Individual with genotype $I^A I^B$ will have both A and B antigens and therefore, AB blood group;

Phenotype of heterozygote different from either homozygote whereby $I^A I^A$ gives A blood group and $I^B I^B$ gives B blood group;

[3]

- (b) In humans, a gene that codes for the production of a protein, called factor VIII, is located on the X chromosome. The dominant allele for this gene produces factor VIII, but the recessive allele does not produce factor VIII.

A person who is unable to make factor VIII has haemophilia in which the blood fails to clot properly.

Explain why a man with haemophilia cannot pass haemophilia to his son but may pass haemophilia to his grandson.

son receives Y chromosome from father / did not inherit X chromosome containing haemophilia allele from father ;

father will pass haemophilia allele to daughter(s) ;

daughter may pass allele to, her son / his grandson ; accept on diagram

[3]

- (c) A gene for feather colour in chickens is carried on an autosome. This gene has two alleles, black (C^B) and splashed-white (C^W). When a male chicken with black feathers is mated with a female chicken with splashed-white feathers, all the offspring have blue feathers. This also occurs when a male chicken with splashed-white feathers is crossed with a female with black feathers.

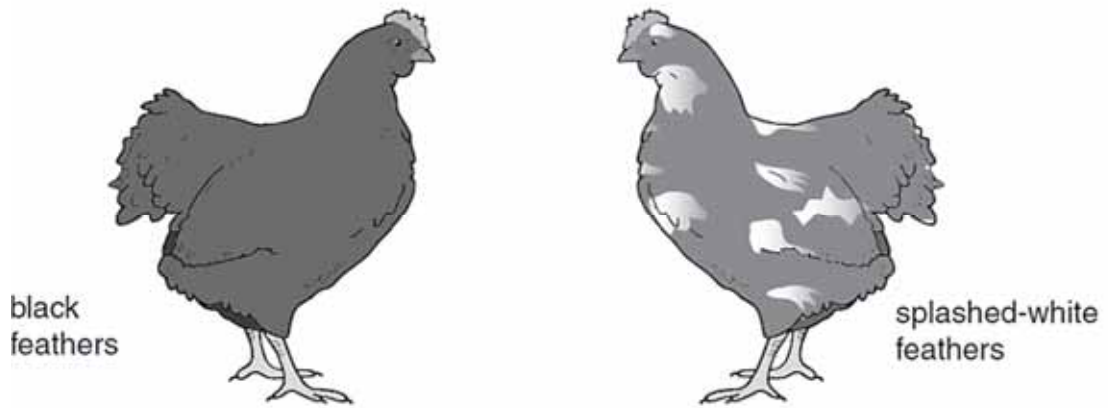


Fig. 7.1

Another gene may cause stripes on feathers (barred feathers). This gene is carried on the X chromosome. The allele for barred feathers (X^A) is dominant to the allele for nonbarred feathers (X^a).

In chickens, the male is homogametic and has two X chromosomes while the female is heterogametic and has one X chromosome and one Y chromosome.

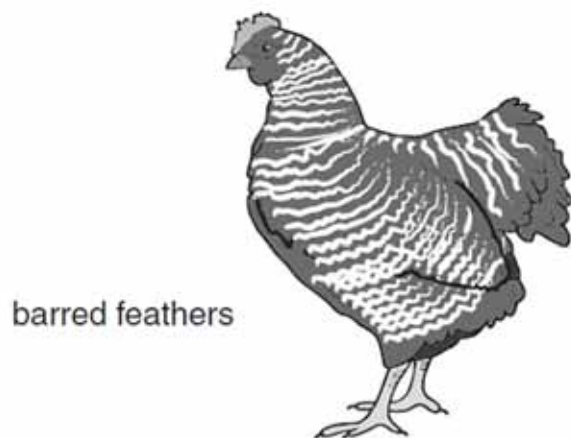


Fig. 7.2

- (i) A male chicken with black, non-barred feathers was crossed with a female chicken with splashed-white, barred feathers. All the offspring had blue feathers, but the males were barred and the females were non-barred.

Using the symbols given above draw a genetic diagram to show this cross.

<i>parents'</i> <i>phenotype</i>	male, black, non-barred feathers.	female, splashed-white, barred feathers.
<i>genotype</i>		
<i>gametes</i>		
<i>offspring</i> <i>genotypes</i>		
<i>phenotypes</i>	male, blue, barred feathers.	female, blue, non-barred feathers.

[5]

(c) (i) (male) $C^B C^B X^a X^a$; x (female) $C^W C^W X^A Y$;
 (gametes) $C^B X^a$ $C^W X^A$ or $C^W Y$;
 $C^B C^W X^A X^a$; $C^B C^W X^a Y$;
 (male, blue, barred) (female, blue, non-barred)

accept other symbols but only with key
 if male XY and female XX then mark gametes and offspring genotypes to max 2
 if other symbols used but no key then mark to max 2

[5]

- 1 mark for parental genotype;
 1 mark for gametes;
 1 mark for offspring genotype and matching;

[3]

- (ii) Explain how a farmer could use a breeding programme to find out the genotype of a male chicken with blue, barred feathers.

with non-barred female ;
 if all offspring barred, must be $X^A X^A$ / homozygous ;
 if some offspring non-barred, must be $X^A X^a$ / heterozygous ;

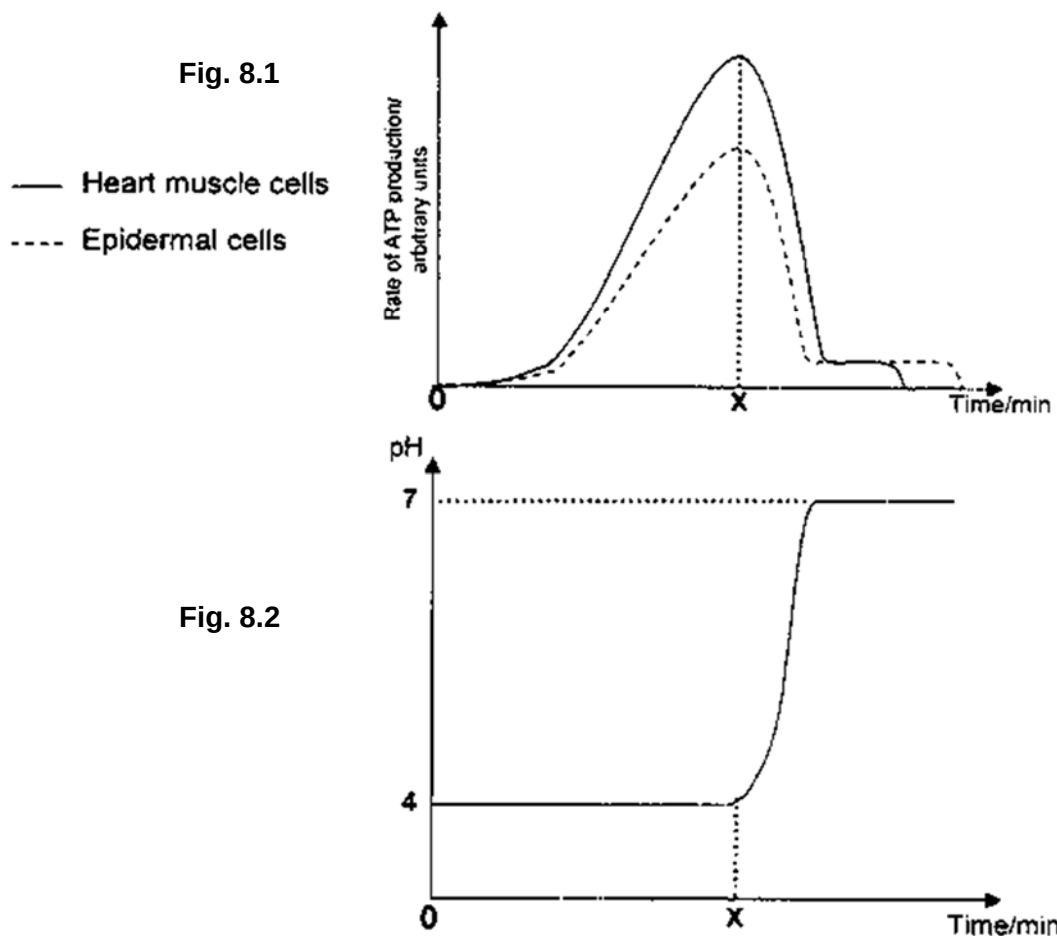
[3]

[Total: 12]

- 8 Heart muscle cells and epidermal cells were extracted from Chinese hamsters. The cells were lysed and the mitochondria and cytosol were isolated. The mitochondria and cytosol were then mixed and re-suspended in a culture of essential nutrients. This suspension system was used to study the process of cellular respiration.

At time 0, glucose was added to the system. At Time X, Digitonin, a detergent which disrupts membranes was introduced to the suspension system. A probe was used to measure the concentrations of ATP as well as the pH level in the mitochondria.

The experimental results are recorded in the graphs shown. **Fig. 8.1** shows the rate of ATP production for heart muscle cells and epidermal cells. **Fig. 8.2** shows the pH level of the mitochondria in both heart muscle and epidermal cells.



- (a) Account for the difference in the level of ATP production in both tissues after glucose was added.

Heart cells produce a higher level of ATP as it contains more mitochondria than epidermal cells;

[1]

- (b) With reference to **Fig. 8.1**, explain the changes in ATP production over time for the heart muscle cell suspension.

2 marks max for the first 3 marking points

High energy electrons from reduced coenzymes are passed down a series of electron carriers on the ETC, each with an energy level lower than the one preceding it;

Energy from the flow of electrons is used to actively pump protons from matrix to intermembrane space, through conformational change of proteins in ETC;

Rate of ATP production increases with time before addition of X due to protons diffuse down the electrochemical proton gradient back into mitochondrial matrix through ATP synthase/ stalked particles, synthesizing ATP from ADP and Pi;

Ref to effect resulting from the membrane damage e.g. no ETC / ATP synthase / electrochemical gradient etc

Initial lag in ATP production because glycolysis is occurring to produce 2 ATP per glucose by substrate level phosphorylation;

OR

Rate of ATP production levels off because glycolysis can still occur and decreases to zero when glucose is used up;

[3]

- (c) With reference to **Fig. 8.2**, state which region of the mitochondrion the pH probe was measuring. Explain your conclusion.

Intermembrane space;

Ref to pH4 + protons are actively pumped from matrix into intermembrane space;

[2]

- (d) Suggest why cytosol was used to re-suspend the mitochondria.

Enzymes involved in glycolysis are present in the cytosol;

[1]

- (e) From your biological knowledge, explain the adaptation of the double membrane for its role in the production of energy.

Membrane is impermeable to protons, creating electrochemical proton gradient across the inner mitochondrial membrane;

Highly folded inner membrane increase surface area for stalked particles containing ATP synthase and electron carriers to be embedded;

Compartmentalisation so that reactions can occur in different locations (ref. to provision of optimal conditions for enzymes such as that for Krebs' Cycle to work);

[2]

[Total: 9]

- 9 The camel family, Camelidae, are well-known for their ability to survive the hot and dry conditions of the desert, but studies have found that they once thrived in colder climates. A recent finding by a group of scientists unearthed fossilised remains of a giant species of camel in North America. This giant species of camel is closely-related to the one-humped Dromedary camel now found in Africa (**Fig. 9.1**). The distinctive humps of both camels are fat stores.

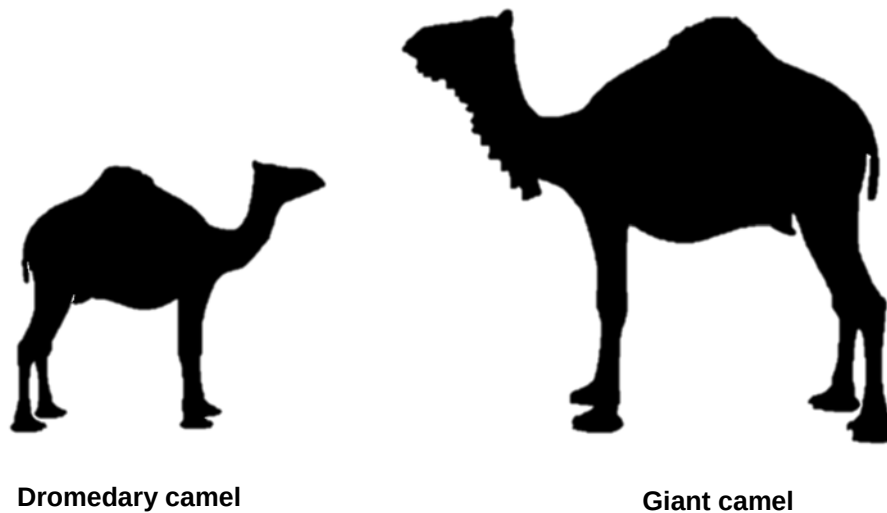


Fig. 9.1

- (a) Suggest how the presence of hump in both the ancestral Giant camel and modern-day Dromedary camel allow them to adapt to their respective habitats.

Ancestral Giant camel in the cold Arctic: Fats in hump used as food store to release energy in the form of ATP / Fats used to keep the camels warm as insulator of heat ;

Dromedary camel in the desert: Fats in hump mainly used to release metabolic water when oxidised in respiration ;

[2]

Until only about two or three million years ago, the Camelids were largely confined within North America. Following the formation of the Bering Land Bridge and the Isthmus of Panama, camels migrated from North America to Asia and South America respectively. Today, camels are no longer found in North America.

Three modern-day groups of species survive today:

- the one-hump Dromedary camel of north Africa and southwest Asia,
- the two-hump Bactrian camel of central Asia and
- the South American Camelids group which has diverged into four species: llamas, alpacas, guanacos, and vicuñas.

Fig. 9.2 shows the distribution of modern-day camels in the world.

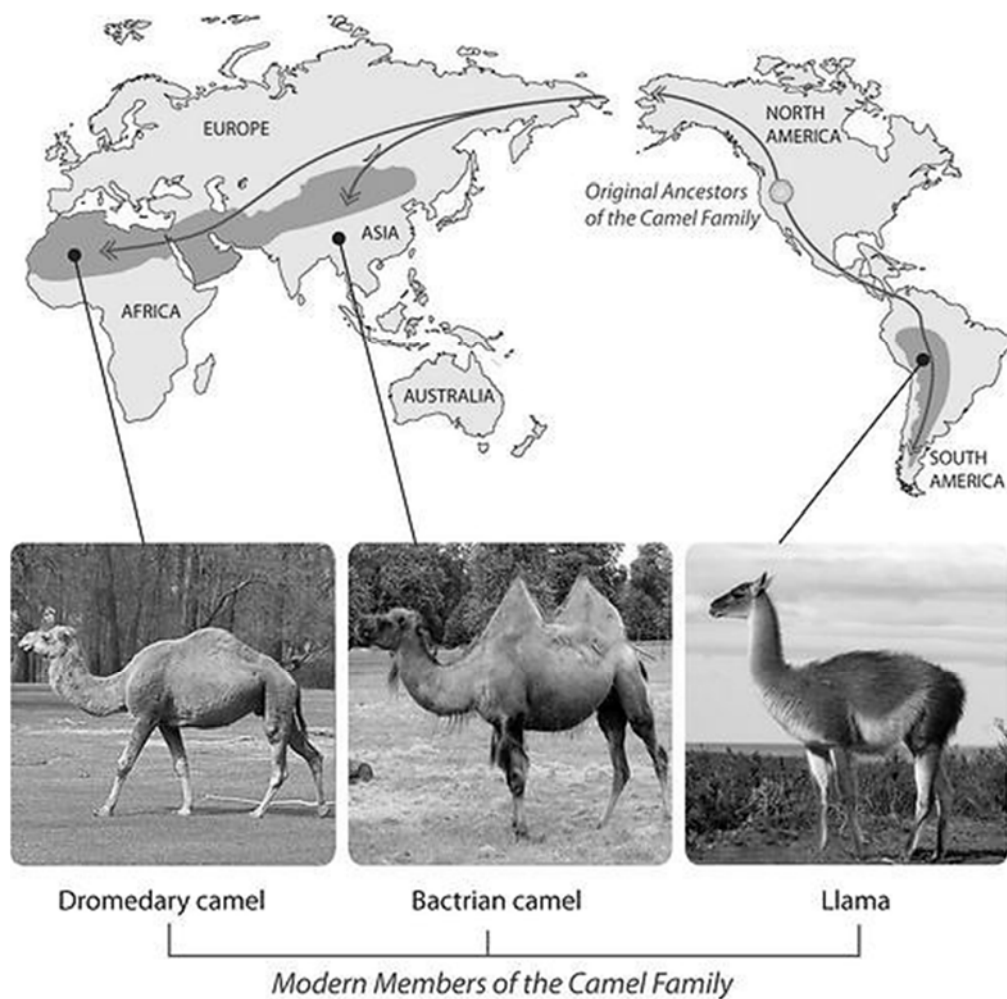


Fig. 9.2

The various species of the modern-day camels evolved from the ancestral population in North America.

- (b) With reference to **Fig. 9.2**, describe how natural selection could have occurred to give rise to the various species of camels in the world today.

Different environments resulting in different selection pressures;

By natural selection, individuals who are better adapted survive till maturity and produce offspring who inherit the beneficial / favourable / advantageous alleles;

As North America and Asia are not continuous/ separated by sea, there is geographical isolation /

There is behavioural / physiological isolation resulting in two populations arising in North America and South America;

Isolation in different continents results in prevention of interbreeding and hence there is no gene flow between sub-populations and over time (and space) evolved into separate species due to accumulation in genetic differences;

when reproductive isolation/ of two or more populations of the same species occurs/ when two populations are no longer able to interbreed to form fertile and viable offspring, new species arise;

2 marks for natural selection + 2 marks for speciation

[4]

- (c) The Arabian camels in Australia typically have sand coat colour. However, there is a small percentage of camels which have albino coat. It is known that the albino coat is a recessive condition and it reduces the life-span of the camel. The population of the sand coat and albino coat camels in a region was documented.

- (i) Explain why population is considered the smallest unit of evolution.

Population refers to a group of interbreeding individuals of the same species;

Evolution is the genetic changes / change in allelic frequency that occur in the population of organisms through time, leading to differences amongst them.

But a population can have variation in traits / characteristics and individual cannot acquire variation;

thus be subjected to forces of natural selection, and can undergo all the changes in genotypes and phenotypes associated with evolutionary change.

[2]

- (ii) Explain whether it is likely that the albino camel will disappear completely from the population over time. Assume there is no introduction of new albino camels into the population.

Unlikely;

Reference to diploidy where recessive allele is protected in the heterozygote condition (i.e. heterozygote protection) ;

When two heterozygote interbreed, there is 25% chance of producing a homozygote recessive camel (gg) showing the albino coat [1/2]

[2]

[Total: 10]

10 (a) The infectious disease tuberculosis (TB) is caused by a bacterium.

Each of the descriptions **A** to **C** describes a cell structure found in prokaryotic cells **and** in plant cells.

For each of the descriptions **A** to **C**:

- name the cell structure described
- state **one** difference in this structure between a prokaryotic cell and a plant cell.

A the site of polypeptide synthesis

cell structure ribosomes

difference 70S ribosomes vs 80S ribosome

B the genetic material of the cell

cell structure DNA/ chromosome

difference circular DNA vs linear DNA / no histone proteins vs histone proteins / not surrounded by nuclear envelope vs surrounded by nuclear envelope

C the structure that provides a rigid shape to the cell and prevents osmotic lysis

cell structure cell wall

difference peptidoglycan cell wall vs cellulose cell wall

[3]

(b) Describe how TB is transmitted from an infected person to an uninfected person.

1 infected person, coughs / sneezes / breathes out/AW, droplets containing, bacteria/ pathogen/ M. tuberculosis ;

2 aerosol / airborne droplets / droplets in air / moist air, inhaled/ inspired/ breathed in (by uninfected person) ;

[2]

(c) Outline how the use of vaccine can give protection against diseases such as tuberculosis.

1 vaccine contains (bacterial) antigen(s);

2 During primary immune response, naïve B-lymphocytes @ B cells are activated and formation of plasma cells that secrete antibody / immunoglobulin (against TB antigens)/ antitoxins;

3 T-helper lymphocytes secrete cytokine and (cytokine) increases humoral response / stimulates cytotoxic T cells / stimulates macrophages;

4 memory B cell production;

- 5 During secondary (immune) response, response on further infection is faster;
 - 6 memory B cell quickly undergo clonal expansion and higher levels of antibodies produced (during further infection);
 - 7 active artificial immunity (against cholera);
 - 8 AVP e.g. idea of specific antibody against each of the different vaccine antigens;
-

[5]

[Total: 10]